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Thermodynamic assessment of the Ga–Lu system by the combination of ab-initio calculations and the CALPHAD approach

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ABSTRACT

First-principles calculations within DFT (density functional theory) were employed to calculate the enthalpies of formation of the metal compounds in the Ga–Lu system. The phase diagram and thermodynamic data of this system were critically optimized using the CALPHAD method and Thermo-Calc software. The Ga–Lu phase diagram contains five intermetallic compounds: Ga_3Lu, Ga_2Lu, Ga_5Lu_3, $GaLu$ and Ga_3Lu_5. These phases were stoichiometric except Ga_3Lu_5 that has a homogeneity range, was modeled by a two-sublattice model with substitution in each sublattice.

The Ga–Lu system was evaluated using the substitutional solution model and the associate model to the description of the liquid phase, in which the excess Gibbs free energy (^{ex}G) was formulated with Redlich–Kister polynomial. The values of the enthalpies of formation that we obtained using two models are in good agreement with those calculated in this work using the VASP code and with experimental data from the literature. The optimization procedure allows us to optimize for the first time all invariant equilibria, thermodynamic properties of the liquid phase and intermetallic compounds of the Ga–Lu system.

1. Introduction

Gallium has many industrial applications; it is used in the form of phosphide (GaP) and arsenide (GaAs) as substrates in integrated digital circuits. It is also used in semiconductor manufacturing for very low-power lighting and the development of photovoltaic panels [1]. On the other hand, Lutetium can be used as a phthalocyanine such as photo-conversion of pollutants [2], non-linear optics [3], compact disc read-write [4] and photodynamic therapy [5]. The isotope of lutetium is also used in nuclear medicine for the treatment of certain neuroendocrine tumors. In addition, rare-earth elements and their alloys are of great interest for the development and design of new materials due to their rich physical properties.

The objective of this paper is to optimize the Ga–Lu binary system using the CALPHAD technique [6], in which the thermodynamic characteristics are analyzed by thermodynamic models for the Gibbs free energy of all phases. In the present work, we have performed and presented the first thermodynamic modeling of the Ga–Lu binary system, which is a continuation of our studies on Gallium with rare-earth elements [7]. The modeling process allows us to calculate the values of the thermodynamic properties, to draw the phase diagram of this system, to complete the experimental data, and to study its consistency with the results optimized by the two models.

2. Review of the literature data

2.1. Phase diagram and crystallographic data

The Ga–Lu binary system is one of the Ga-REE (REE = Rare Earth Elements) binary systems studied by Yatsenko et al. [8] using different experimental methods such as X-ray diffraction and differential thermal analysis. These authors noted the existence of five intermetallic compounds: Ga_3Lu, Ga_2Lu, Ga_3Lu_2, $GaLu$ and Ga_3Lu_5. Where the compounds Ga_3Lu_2, $GaLu$ and Ga_3Lu_5 melt congruently at 1280 °C, 1350 °C and 1420 °C, respectively. Ga_3Lu, and Ga_2Lu were formed by a peritectic reactions at 930 °C and 1170 °C, respectively. Four eutectics reactions

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Table 1
Summary of crystal structure data of all compounds in the Ga–Lu binary system.

Phase	Composition, at.% Lu	Space group	Structure type	Lattice parameters (Å)			References
				a	b	c	
αGa	0.00	*Cmca*	*Ga*	4.524	7.661	4.519	[11]
				4.501	7.748	4.624	This work
Ga_3Lu	25.0	$Pm\overline{3}m$	Cu_3Au	4.169	4.169	4.169	[12]
				4.191	4.191	4.191	[13]
				4.200	4.200	4.200	[14]
				4.218	4.218	4.218	This work
Ga_2Lu	33.3	*Imma*	Cu_2Ce	4.185	6.85	8.044	[15]
				4.208	6.856	8.133	This work
Ga_5Lu_3	37.5	*Pnma*	Ga_5Tm_3	11.24	9.515	5.96	[9]
				11.320	9.561	5.965	This work
GaLu	50.0	*Cmcm*	*CrB*	4.209	10.581	4.000	[16]
				4.273	10.637	3.977	This work
Ga_3Lu_5	62.5	$P6_3/mcm$	Si_3Mn_5	8.363	8.363	6.275	[17]
				8.382	8.382	6.341	[18]
				8.402	8.402	6.281	[19]
				8.414	8.414	6.337	This work
αLu	100	$P6_3/mmc$	*Mg*	3.518	3.518	5.581	[20]
				3.526	3.526	5.471	This work

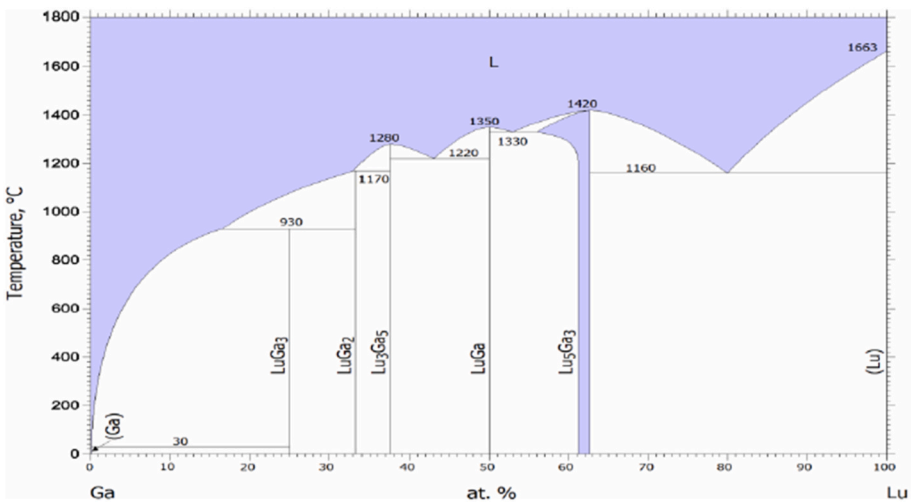

Fig. 1. Phase diagram of the Gallium–Lutetium system given by Ref. [10].

were formed in the $Ga-Lu$ system: between (α Ga) and Ga_3Lu, the eutectic is degenerate, between Ga_3Lu_2 and *GaLu* at 1220 °C and about 43 at.% Lu, between *GaLu* and Ga_3Lu_5 at 1330 °C and about 53 at.% Lu; between Ga_3Lu_5 and the $\alpha-$ modification of Lu at 1160 °C and about 80 at.% Lu. However, Yatsenko et al. [9] reported the existence of a new Ga_5Lu_3 phase using single-crystal X-ray methods, with a Ga_5Tm_3 crystal structure, and Pnma space group (Table 1).

In 1990, Massalski [10] also provided an updated version of the phase diagram of the $Ga-Lu$ system (Fig. 1) based on the work of Yasenko et al. [8,9]. This phase equilibria contains five intermediate phases: Ga_3Lu, Ga_2Lu, Ga_5Lu_3, *GaLu* and Ga_3Lu_5 and two solid terminal solutions, (αGa) and (αLu), this diagram is a little different to that of [8] in the Ga-rich side, since there is a not degenerate eutectic reaction at 0.001 at.% Lu and 29.77 °C.

The crystallographic data of the pure elements and five intermediate compounds formed in the Ga–Lu system are given in Table 1.

2.2. Thermodynamic data

Bayanov and Soboleva [21] determined by the EMF method (at 673–875 K) the enthalpy of formation and entropy of the Ga_3Lu intermetallic compound twice, the first time in Ga (liquid) and Lu (solid), the second time in Ga (solid) and Lu (solid). Based on their calorimetric measurement at 1373 ± 2 K, Meshel and Kleppa [22] measured the standard enthalpy of formation of the same compound and found the value of -52.2 ± 2.3 kJ/mol. Most of the enthalpies of formation of binary compounds have been calculated by ab-initio method and presented in online databases such as: Materials project [23], OQMD [24] and Jarvis [25]. In addition, Miedema [26] also computed the enthalpies of formation of all compounds formed in the Ga–Lu system by his model.

3. First-principles calculations

According to the analysis in Section 2.2, there is only one enthalpy of formation value for the Ga_3Lu compound [21,22], then there is no thermodynamic information exists for the other intermetallic compounds (Ga_2Lu, Ga_5Lu_3, *GaLu* and Ga_3Lu_5).

As illustrated in Table 1, and in order to clarify the discrepancies between the literature data, all crystal structures of five compounds are well optimized, which leads us to start the first-principles calculations of the enthalpies of formation.

The total energies of all intermetallic compounds in the Ga–Lu system, as well as those of pure elements in the ground state (0 K), were calculated using the density functional theory (DFT) [27], within the generalized gradient approximation (GGA) [28] and using the projector augmented wave (PAW) [29] pseudopotentials for treatment of

Table 2
Summary of enthalpies of formation (ΔH) and entropies of formation (ΔS) for all the compounds in the Ga–Lu binary system according to the present work.

Compound	$\Delta H(kJ/mol)$	$\Delta S(J.K^{-1}mol^{-1}at^{-1})$	Method used	References
Ga_3Lu	− 49.4	–	Miedema's model	[26]
	− 52.2	–	Calorimetry	[21]
	− 45.2	6.9	EMF	[22]
	− 48.240	–	Ab-initio[a]	[23]
	− 46.599	–	Ab-initio[a]	[24]
	− 45.644	–	Ab-initio[a]	[25]
	− 47.564	–	Ab-initio	This work
	− 47.551	4.54	*Calphad* [R–K]	Current work
	− 47.254	7.22	*Calphad* [A]	Current work
Ga_2Lu	− 62.5	–	Miedema's model	[26]
	− 55.476	–	Ab-initio[a]	[23]
	− 54.247	–	Ab-initio[a]	[25]
	− 55.572	–	Ab-initio	This work
	− 58.860	2.93	*Calphad* [R–K]	Current work
	− 56.323	7.90	*Calphad* [A]	Current work
Ga_5Lu_3	− 69.4	–	Miedema's model	[26]
	− 58.370	–	Ab-initio[a]	[23]
	− 58.466	–	Ab-initio	This work
	− 57.084	6.13	*Calphad* [R–K]	Current work
	− 58.614	8.46	*Calphad* [A]	Current work
$GaLu$	− 71.9	–	Miedema's model	[26]
	− 58.756	–	Ab-initio[a]	[23]
	− 58.563	–	Ab-initio[a]	[24]
	− 59.394	–	Ab-initio[a]	[25]
	− 58.659	–	Ab-initio	This work
	− 52.406	9.51	*Calphad* [R–K]	Current work
	− 58.283	7.64	*Calphad* [A]	Current work
Ga_3Lu_5	− 62.1	–	Miedema's model	[26]
	− 46.889	–	Ab-initio[a]	[23]
	− 46.503	–	Ab-initio[a]	[24]
	− 48.643	–	Ab-initio[a]	[25]
	− 47.854	–	Ab-initio	This work
	− 48.128	7.54	*Calphad* [R–K]	Current work
	− 48.004	6.32	*Calphad* [A]	Current work

[R–K]Redlich–Kister model [34] and [A]Associate model [36].
[a] Ab-initio calculation data obtained from online databases: Materials project [23], OQMD [24], and Jarvis [25].

core-electron interactions, as implemented in the Vienna Ab-initio Simulation Package (VASP) [30,31]. We used PAW potentials with 13 (3d10 4s2 4p1) valence electrons for Ga and with 9 (5p6 6s2 5d1) valence electrons for Lu. The kinetic cut-off energy was set to 520 eV, and the Gamma-centered k-mesh [32] for the Brillouin zone integrations was used. The maximum distance between two k-points in reciprocal space was equal to 0.23 2*PI/Angstrom. Such parameters were sufficient to allow the convergence of formation enthalpies of compounds at a level better than 1 meV/atom.

The enthalpy of formation of an intermetallic compound at 0 K can be calculated as the difference between the total energy of that compound and the total energies of its constituent elements in their stable states from the following formula:

$$\Delta_f H = E_{total}^{Ga_xLu_y} - \frac{x}{x+y} \times E_{total}^{Ga-Ortho} - \frac{y}{x+y} \times E_{total}^{Lu-Hcp} \quad (1)$$

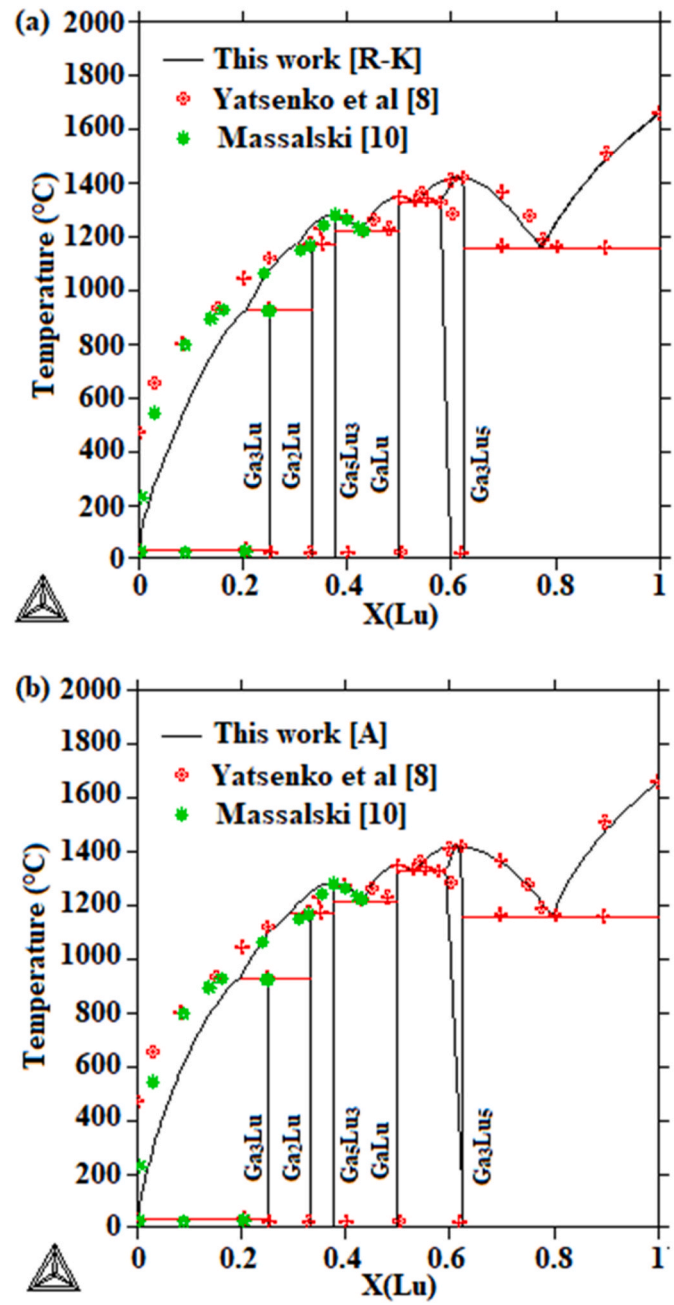

Fig. 2. Ga–Lu phase diagram calculated, compared with the experimental data from the literature [8,10]. Using **(a)** Redlich–Kister model [34] and **(b)** the associate model [36].

Where $E_{total}^{Ga_xLu_y}$ is the total energy of the Ga_xLu_y compound. $E_{total}^{Ga-Ortho}$ and E_{total}^{Lu-Hcp} were the total energies of the pure elements Ga and Lu in their most stable structures, respectively.

The total energies of the compounds and their constituting elements in their ground states were calculated from interpolation of energy-volume curves with cubic splines. The energy-volume curves were obtained from cell shape and atomic positions relaxations at different fixed volumes.

As the influence of the pressure term on the condensed phases was ignored and the total energy was calculated at 0 K, the energy of formation calculated in this paper was taken as the enthalpy of formation at 298 K. The calculated values should be compared with the experimental results.

Table 3
Summary of the thermodynamic parameters calculated in the Ga–Lu system using Redlich–Kister equation[a].

phases	Thermodynamic models	Parameters
Liquid	$(Ga,Lu)_1$	${}^0L^{Liq}_{Ga,Lu} = -183121 - 4.59*T$
		${}^1L^{Liq}_{Ga,Lu} = -101996 + 54.54*T$
		${}^2L^{Liq}_{Ga,Lu} = 87151 - 22.04*T$
ORT	(Ga)	No excess term optimized
Hcp_A3	(Ga,Lu)	No excess term optimized
Ga_3Lu	$(Ga)_{0.75}:(Lu)_{0.25}$	$G^{Ga_3Lu}_{Ga,Lu} - 0.75*{}^0G^{Ort}_{Ga} - 0.25*{}^0G^{hcp}_{Lu} = -47551 - 4.54*T$
Ga_2Lu	$(Ga)_{0.667}:(Lu)_{0.333}$	$G^{Ga_2Lu}_{Ga,Lu} - 0.667*{}^0G^{Ort}_{Ga} - 0.333*{}^0G^{hcp}_{Lu} = -58860 - 2.93*T$
Ga_5Lu_3	$(Ga)_{0.625}:(Lu)_{0.375}$	$G^{Ga_5Lu_3}_{Ga,Lu} - 0.625*{}^0G^{Ort}_{Ga} - 0.375*{}^0G^{hcp}_{Lu} = -57084 - 6.13*T$
$GaLu$	$(Ga)_{0.5}:(Lu)_{0.5}$	$G^{GaLu}_{Ga,Lu} - 0.5*{}^0G^{Ort}_{Ga} - 0.5*{}^0G^{hcp}_{Lu} = -52406 - 9.51*T$
Ga_3Lu_5	$(Ga)_{0.375}:(Lu)_{0.625}$	$G^{Ga_3Lu_5}_{Ga,Lu} - 0.375*{}^0G^{Ort}_{Ga} - 0.625*{}^0G^{hcp}_{Lu} = -48128 - 7.54*T$
		$G^{Ga_3Lu_5}_{Ga,Ga} - H^{Hcp}_{Ga} = 25000 + GHSERGa$
		${}^0L^{Ga_3Lu_5}_{Ga:Ga,Lu} = -100000 - 7*T$

[a] In J/mol-atom.

Table 4
Summary of the thermodynamic parameters calculated in the Ga–Lu system using the associate model[a].

phases	Thermodynamic models	Parameters
Liquid	$(Ga,Lu,GaLu)_1$	$L^{Liq}_{GaLu} = -101918 - 9.71*T$
		${}^0L^{Liq}_{Ga,GaLu} = -34370 - 19.49*T$
		${}^1L^{Liq}_{Ga,GaLu} = -54020 + 23.56*T$
		${}^0L^{Liq}_{GaLu,Lu} = -32880 + 18.17*T$
		${}^1L^{Liq}_{GaLu,Lu} = 22011 - 12.01*T$
ORT	(Ga)	No excess term optimized
Hcp_A3	(Ga,Lu)	No excess term optimized
Ga_3Lu	$(Ga)_{0.75}:(Lu)_{0.25}$	$G^{Ga_3Lu}_{Ga,Lu} - 0.75*{}^0G^{Ort}_{Ga} - 0.25*{}^0G^{hcp}_{Lu} = -47254 - 7.22*T$
Ga_2Lu	$(Ga)_{0.667}:(Lu)_{0.333}$	$G^{Ga_2Lu}_{Ga,Lu} - 0.667*{}^0G^{Ort}_{Ga} - 0.333*{}^0G^{hcp}_{Lu} = -56323 - 7.90*T$
Ga_5Lu_3	$(Ga)_{0.625}:(Lu)_{0.375}$	$G^{Ga_5Lu_3}_{Ga,Lu} - 0.625*{}^0G^{Ort}_{Ga} - 0.375*{}^0G^{hcp}_{Lu} = -58614 - 8.46*T$
$GaLu$	$(Ga)_{0.5}:(Lu)_{0.5}$	$G^{GaLu}_{Ga,Lu} - 0.5*{}^0G^{Ort}_{Ga} - 0.5*{}^0G^{hcp}_{Lu} = -58283 - 7.64*T$
Ga_3Lu_5	$(Ga)_{0.375}:(Lu)_{0.625}$	$G^{Ga_3Lu_5}_{Ga,Lu} - 0.375*{}^0G^{Ort}_{Ga} - 0.625*{}^0G^{hcp}_{Lu} = -48004 - 6.32*T$
		$G^{Ga_3Lu_5}_{Ga,Ga} - H^{Hcp}_{Ga} = 5000 + GHSERGa$
		${}^0L^{Ga_3Lu_5}_{Ga:Ga,Lu} = -90000 - 10*T$
		${}^1L^{Ga_3Lu_5}_{Ga:Ga,Lu} = 2903$

[a] In J/mol-atom.

4. Thermodynamic modeling

4.1. Thermodynamic description of unary phases

The Gibbs energy function $G^{\varphi}_i(T) = {}^0G^{SER}_i(T) - {}^0H^{SER}_i(298.15\,K)$ for a pure element i (i = Ga, Lu) in the φ phase is given according to the SGTE [33] by the following formula:

$$G^{\varphi}_i(T) = a + b \times T + c \times T \times ln(T) + d \times T^2 + e \times T^3 + f \times T^{-1} + g \times T^7 + h \times T^{-9} \tag{2}$$

Where ${}^0H^{SER}_i(298.15\,K)$ is the molar enthalpy of element i at 298.15 K in its standard element reference (SER), Orthorhombic for Gallium and Hcp for Lutetium. The Gibbs energy $G^{\varphi}_i(T)$ of element i, in its SER state is denoted by GHSERi as follows:

$$GHSER_{Ga} = {}^0G^{Orth}_{Ga} = G^{Orth}_{Ga} - H^{Orth}_{Ga} \tag{3}$$

$$GHSER_{Lu} = {}^0G^{Hcp_A3}_{Lu} = G^{Hcp_A3}_{Lu} - H^{Hcp_A3}_{Lu} \tag{4}$$

4.2. Liquid phase

4.2.1. Substitutional solution model

In this work, at first, the liquid phase is described with a substitutional solution model using the Redlich-Kister polynomial [34]. The Gibbs energy of the liquid phase is expressed as follows:

$$G^{Liq}_m = {}^{ref}G^{Liq} + {}^{id}G^{Liq} + {}^{exc}G^{Liq} \tag{5}$$

Where ${}^{ref}G^{Liq}$ is the reference part, ${}^{id}G^{Liq}$ the ideal part and ${}^{exc}G^{Liq}$ the excess part.

As used in the thermo-calc software [35]:

$${}^{ref}G^{Liq}(T) = \left({}^0G^{Liq}_{Ga}(T) - H^{SER}_{Ga}\right)x_{Ga} + \left({}^0G^{Liq}_{Lu}(T) - H^{SER}_{Lu}\right)x_{Lu} \tag{6}$$

$${}^{id}G^{Liq} = RT(x_{Ga} ln\, x_{Ga} + x_{Lu} ln\, x_{Lu}) \tag{7}$$

With R is the gas constant, T the temperature, x_{Ga} and x_{Lu} are the fraction of elements Ga and Lu, respectively.

The excess Gibbs energy ${}^{exc}G^{Liq}_m(T)$ terms of all the solution phases was modeled by the Redlich-Kister [34] formula:

Table 5
Summary of invariant equilibria in the Ga–Lu system.

Invariant reactions	Liquid concentration *x(Lu)/at*			Temperature *T(K)*		
	Measured	Calculated (this work)		Measured	Calculated (this work)	
		[R–K]	[A]		[R–K]	[A]
$Liq \rightarrow (\alpha Ga)$	0.000 [10]	0.000	0.000	302 [10]	302	302
$Liq \rightarrow Ga_3Lu + (\alpha Ga)$	0.001 [10]	0.0009	0.001	302 [10]	302.8	302.8
$Liq + Ga_2Lu \rightarrow Ga_3Lu$	0.150 [10]	0.200	0.190	1203 [10]	1203	1203
$Liq + Ga_5Lu_3 \rightarrow Ga_2Lu$	0.320 [10]	0.302	0.293	1443 [10]	1444	1443
$Liq \rightarrow Ga_5Lu_3$	0.375 [10]	0.375	0.375	1553 [10]	1553	1555
$Liq \rightarrow GaLu + Ga_5Lu_3$	0.433 [10]	0.427	0.432	1493 [10]	1494	1490
$Liq \rightarrow GaLu$	0.500 [10]	0.500	0.500	1623 [10]	1622	1623
$Liq \rightarrow Ga_3Lu_5 + GaLu$	0.530 [10]	0.528	0.532	1603 [10]	1604	1601
$Liq \rightarrow Ga_3Lu_5$	0.625 [10]	0.625	0.625	1693 [10]	1693	1692
$Liq \rightarrow Ga_3Lu_5 + (\beta Lu)$	0.800 [10]	0.772	0.795	1433 [10]	1431	1432
$Liq \rightarrow (\beta Lu)$	1.000 [10]	1.000	1.000	1936 [10]	1936	1936

[R–K]Redlich–Kister model [34] and [A]Associate model [36].

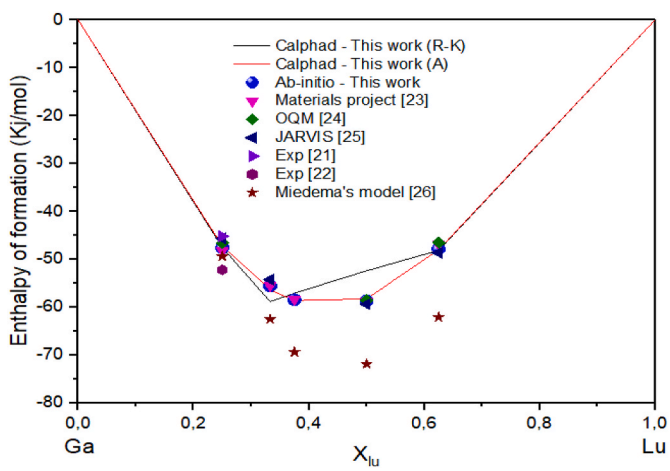

Fig. 3. Calculated enthalpies of formation of solid alloys and intermetallic compounds in the Ga–Lu system. ([R–K]Redlich–Kister model [34] and [A]Associate model [36]).

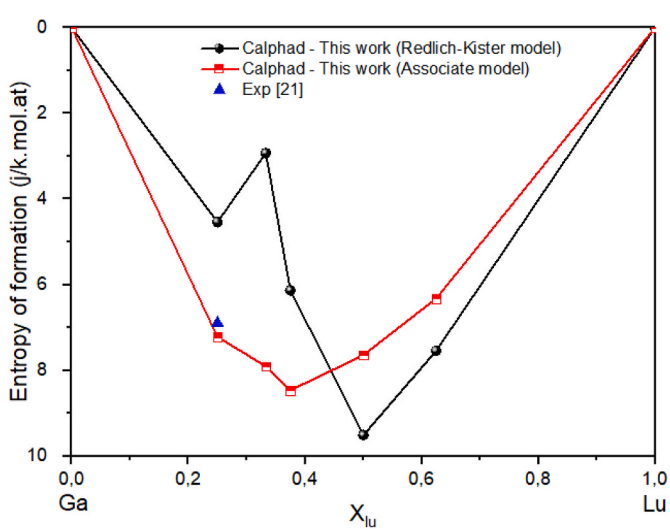

Fig. 4. Calculated entropies of formation of intermetallic compounds in the Ga–Lu system compared with the literature data [21].

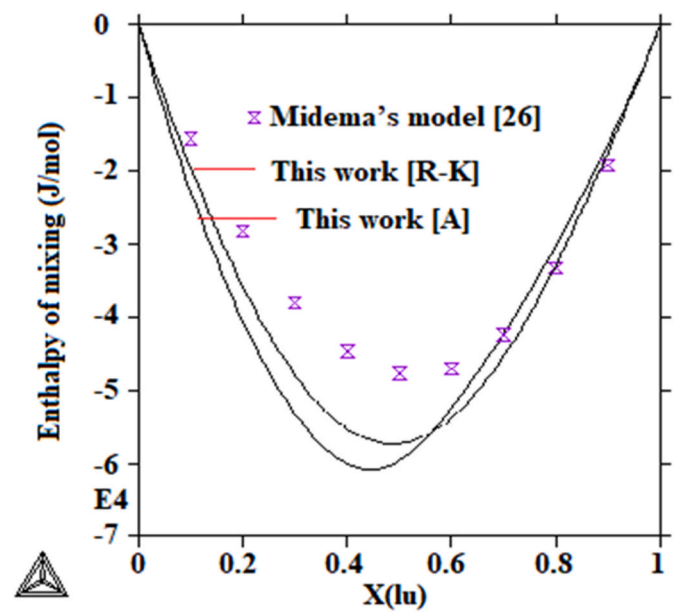

Fig. 5. Enthalpy of mixing for the liquid phase calculated at 1600 K compared with Midema's model [26]. ([R–K]Redlich–Kister model [34] and [A]Associate model [36]).

$$^{exc}G_m^{Liq}(T) = x_{Ga}x_{Lu}\sum_i {}^iL_{Ga,Lu}^{Liq}(x_{Ga} - x_{Lu})^i \tag{8}$$

Where ${}^iL_{Ga,Lu}^{Liq}(T)$ is the binary interaction parameter of the two constituents Ga and Lu in the liquid phase, which to be evaluated in this work. With a_i and b_i are the coefficients to be optimized.

$$^iL_{Ga,Lu}^{Liq}(T) = a_i + b_iT \qquad (i = 0,\ 1,\ 2...) \tag{9}$$

With the excess entropy $s = -b_i$.

4.2.2. Associate model

In addition to the linear dependence model, we also used the associate model for the description of the liquid phase in the Ga–Lu system. This system has strong attractive interactions. Therefore, it is likely that the Ga–Lu system exhibits short-range ordering. The association model was developed by Sommer et al. [36], it was chosen in this work for the description of the liquid phase with the constituent species Ga, Lu and *GaLu* as associates in the liquid. The Gibbs energy of the liquid phase is written as follows:

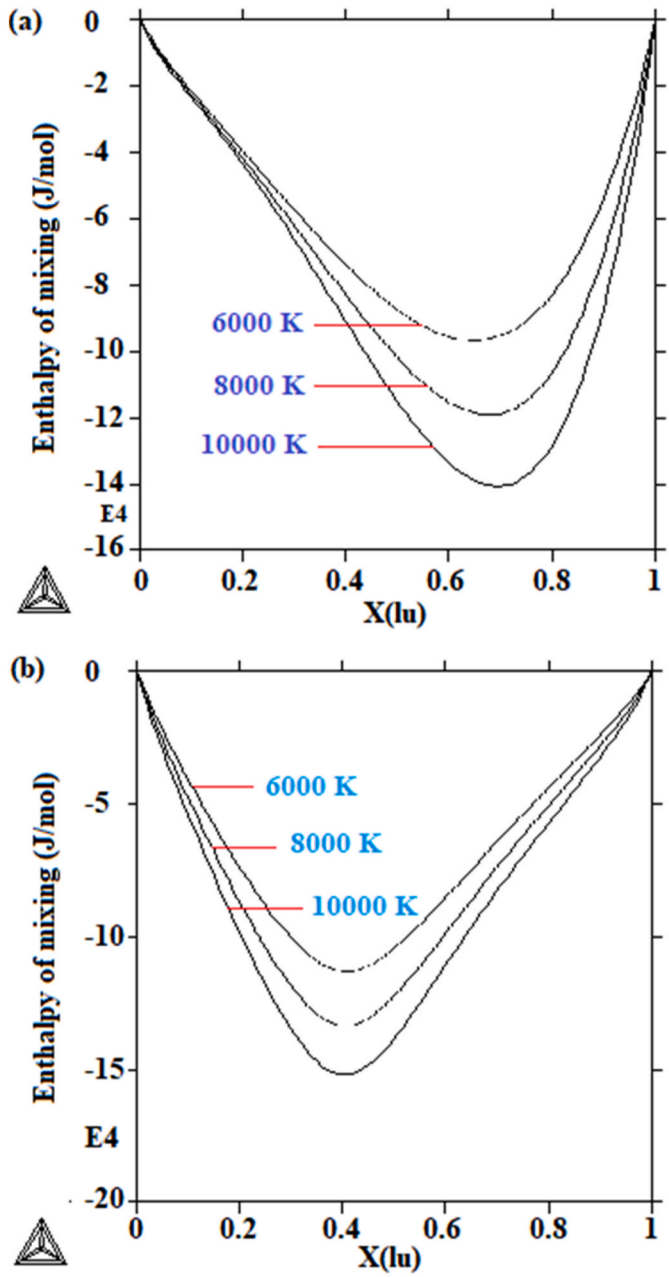

Fig. 6. Calculated enthalpy of mixing of the liquid phase at different temperatures (6000 K–10000 K). using **(a)** Redlich–Kister model [34] and **(b)** the associate model [36].

$$G^{Liq} = \sum_{i=Ga,Lu,GaLu} x_i G_i^{Liq} + RT \sum_{i=Ga,Lu,GaLu} x_i \ln(x_i) + G_{exc}^{Liq} \tag{10}$$

Where x_i is the mole fraction of each species i (i = Ga, Lu and GaLu) in the liquid phase. G_{exc}^{Liq} is the excess term of the Gibbs free energy of the liquid phase which will be described as follows:

$$G_{exc}^{Liq} = \sum_i \sum_{i \neq j} x_i x_j \sum_{v=0}^{n} {}^{v}\mathrm{L}_{i,j}^{Liq} \left(x_i - x_j\right)^v \tag{11}$$

Where ${}^{v}\mathrm{L}_{i,j}^{Liq}$ is the binary interaction parameter, it is described by a function of the linear dependence of temperature by the following equation:

$${}^{v}\mathrm{L}_{i,j}^{Liq} = {}^{v}a_{i,j}^{Liq} + {}^{v}b_{i,j}^{Liq} \times T \tag{12}$$

4.3. Thermodynamic description of stoichiometric compounds

In the Ga–Lu system, there are four stoichiometric compounds: Ga_3Lu, Ga_2Lu, Ga_5Lu_3, and $GaLu$. The Gibbs energy of the Ga_mLu_n compound is expressed by:

$${}^{0}G_{Ga_mLu_n} = \frac{m}{m+n}\,{}^{0}G_{Ga} + \frac{n}{m+n}\,{}^{0}G_{Lu} + a + bT \tag{13}$$

Where ${}^{0}G_{Ga}$ and ${}^{0}G_{Lu}$ are the Gibbs energy of the pure elements Ga and Lu, respectively, a and b are variables to be calculated for each compound of the Ga–Lu system.

4.4. Intermediate phase with homogeneity range

The Ga_3Lu_5 compound is a non-stoichiometric phase with a homogeneity range on the Ga-rich side of strict stoichiometry Ga_3Lu_5. This compound was treated as $(Ga)_{0.375} : (Ga, Lu)_{0.625}$ by a two-sublattice model [37,38]. The Gibbs energy function per mole of the formula unit is given by the following equation:

$$\begin{aligned} {}^{0}G_m^{Ga_3Lu_5} = {} & x_{Ga}\,{}^{0}G_{Ga:Lu}^{Ga_3Lu_5} + x_{Lu}\,{}^{0}G_{Ga:Lu}^{Ga_3Lu_5} + RT(x_{Ga}\ln x_{Ga} + x_{Lu}\ln x_{Lu}) \\ & + x_{Ga}x_{Lu}\sum_i {}^{v}L_{Ga:Ga,Lu}(x_{Ga} - x_{Lu})^v \end{aligned} \tag{14}$$

With x_{Ga} and x_{Lu} are the site part of Ga and Lu constituents located on two sublattices, and the term ${}^{0}G_{Ga:Lu}^{Ga_3Lu_5}$ describes the Gibbs energy of the constituent when the two sublattices are occupied by elements Ga or Lu. ${}^{v}L_{Ga:Lu}$ is the interaction parameter between the Ga and Lu elements in the first or second sublattice, when the other sublattice is occupied by Ga or Lu.

4.5. Optimization strategy

The thermodynamic optimization procedure of the Gibbs free energy function by applying the CALPHAD approach [6] was carried out using the PARROT module of Thermo-Calc software [28]. This module functions by minimizing the sum of the squares of the difference between the input and output values. The optimization was carried out step by step. Firstly, the enthalpies of formation of intermetallic compounds calculated in this work by the VASP code and the experimental data from the literature cited above are introduced as starting values of the coefficient a in equation (13) for each phase. Next, the coefficient b of equation (13) is introduced for each compound. Thirdly, the enthalpy and entropy of the liquid phase were evaluated based on the phase diagram and the thermodynamic data of Miedema [26] associated with the liquid phase.

A consistent set of calculated thermodynamic model parameters is obtained and each input value (experimental or calculated by VASP) is carefully weighted according to its merits, and modified by trial and error during optimization, to achieve satisfactory agreement with the experimental or theoretical data within the expected uncertainty.

5. Results and discussion

Before starting the thermodynamic modeling of the Ga–Lu system by the CALPHAD method, we studied all the stable intermetallic compounds of the known structure in the Ga–Lu system. The lattice parameters and enthalpies of formation calculated by the Ab-initio method in this work are shown in Tables 1–2 and Fig. 3.

The calculated phase diagram of the Ga–Lu binary system obtained in this work using the Redlich-Kister model and the associate model is presented in Fig. 2, compared with experimental data found in the literature [8,10]. The associate model [36] gives a better description of this system to make the liquidus more symmetric with respect to the three congruent compounds compared to the Redlich-Kister model [34].

As can be seen in Table 5, the calculated values of compositions and

temperature by both models for the five intermetallic compounds are in good agreement with those found experimentally in the literature [8,10], especially the one calculated by the associate model [36]. Three congruent, two peritectic and four eutectic reactions of the Ga–Lu binary system are well reproduced taking into account the errors given by Ref. [8] concerning the temperatures, and the compositions of the reactions when plotting manually. However, a disagreement with the experimental phase diagram given by Refs. [8,10] is noted for the proposed liquidus in the Ga-rich part and around the congruent compounds. Since the liquidus in these studies [8,10] is clearly asymmetric around these compounds which is thermodynamically improbable. The current thermodynamic calculation can be considered a more satisfactory prediction; in particular, it corrected the liquidus form in the range from 0 to 30 at.% Lu. More experimental work on this phase diagram might be needed to confirm these results.

Except the compound Ga_3Lu, we did not find any experimental studies regarding the enthalpies of formation and entropies of the compounds: Ga_2Lu, Ga_5Lu_3, $GaLu$ and Ga_3Lu_5 in literature. So, the recourse to Ab-Initio calculation makes the objective of this work. The thermodynamic parameters calculated by the ab-initio method and then optimized by the CALPHAD approach [6] for each phase of the Ga–Lu system using both models are presented in Tables 3 and 4. Fig. 3 shows the curve of enthalpies of formation of intermetallic compounds calculated by the two models compared with our results obtained by the ab-initio method, with experimental data from the literature [21,22], the values from online databases (Materials project [23], OQMD [24] and Jarvis [25]), as well as the values estimated by the Midema's model [26].

Table 2 summarizes the enthalpies of formation and entropies obtained by the CALPHAD modeling in this work for all phases of the Ga–Lu system. As can be seen, a satisfactory agreement is noted between the optimized values and those calculated by ab-initio and with literature data. The optimized enthalpies of formation are acceptable in comparison with the predicted values [26], especially considering the large inaccuracies of Miedema's model. Fig. 4 shows the calculated entropies of formation for all binary compounds of the Ga–Lu system using both models. A good agreement was observed between the calculated value for the Ga_3Lu compound and that measured by Ref. [15].

Fig. 5 shows the enthalpy of mixing of the liquid phase calculated at 1600 K, demonstrating good agreement between the values calculated by the two models and the values predicted by Midema's model [26], especially in the Lu-rich part. To test the stability of the liquid phase at high temperatures we have plotted Fig. 6a and b in the range of 6000 K–10000 K, for the Ga–Lu system. These figures show that the enthalpy of mixing of the liquid phase becomes less negative with increasing temperature. At high temperatures, the liquidus is very stable, and consequently no spinodal points were found up to a temperature of 10 000 K using both models. As indicated in Refs. [39,40,41], we noted that no reverse miscibility gap formation was detected at very high temperatures.

6. Conclusion

The thermodynamic properties and phase diagram of the Ga–Lu system were optimized by combination of first-principles calculations and CALPHAD method. A logical set of thermodynamic parameters was derived based on the literature's reliable data and the Ab-initio calculated results. The assessed phase equilibria are in good agreement with the experimental data, except the liquidus in the Ga-rich part which degenerates a little. This is caused by the asymmetry of the liquidus around the Ga_2Lu compound in the experimental study. In the present work, we determined for the first time the enthalpies of formation and entropies of the intermetallic compounds of the Ga–Lu system, namely: Ga_3Lu, Ga_2Lu, Ga_5Lu_3, $GaLu$ and Ga_3Lu_5 and that of the liquid phase, which allowed us to calculate the phase diagram of this system. Based on the results obtained in this work we recommend an in-depth experimental study to confirm our results and especially the Ga-rich composition range.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2022.102464.

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